

ActInf Textbook Group ~ Cohort 4 ~ Meeting 17 (Chapter 8 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 17 (Chapter 8 part 1) | 54:44

all right

[Music]

welcome it's October 17th 2023 and we're

in our first discussion of chapter 8 in

cohort 4 so to begin with what are just

any thoughts or

memories that you fellows have about

chapter

8 just any any piece that you remember

or anything about

eight um I was curious about the part

where they talk about how movement and

attenuation like sensory attenuation

that was interesting to me where they I

hope I remember calling this correctly

but it was like you didn't want to

perfectly predict it at a certain point

for

like to perceive movement which was

interesting um I don't know if I

perfectly understand that but uh I kind

of see an intuitive aspect to that I

guess oh yeah that's a very interesting

section Andrew anything that that

you um think about yeah yeah um I guess

I'd second an interest in sensory

attenuation as well because I I guess um

we could get to that but like box 8.1 is

kind of the key bit of interest there

the relationship between Precision

attention and sensory attenuation so

they seem to more or less equate

precision and

attention um I guess the idea is um you

have some kind of belief that is you

know I am not moving and you need you
need to downplay or decrease the
Precision of that
belief such that you can actually move
in the first place otherwise if you have
very high Precision I am not
moving um that kind of stays in place
and so you never actually get started on
moving um yeah know I thought that was
really interesting and it comes up a
little bit more later in the chapter on
how that kind of plays out and the
movement of Limbs and such so I thought
that was pretty pretty
fascinating yeah totally and like where
else does that apply like can we
understand the example they give with a
motor

Behavior connected to the Octan fonty
and then what other
analogous um sensory attenuation cases
might be happening let's just see if
it's been explored in a
question

okay we'll definitely look at this
question about
precision and then here's the Box

8.1

okay no
answer what what what's interesting
about
it um a or what what would you want to
know about it or what do I mean um I
guess like I was just it was just
something where
um when I read it it
was it
made uh sense but I'm still trying

to let me see if I can find
it it made sense it was just
um interesting
that we aren't like was it were they
saying that the perception of movement
itself like in order to perceive
movement it was like we were decreasing
how um well we were
predicting uh those States
yeah in order to generate Movement we
must be able to ignore the sensory
consequences which is to say dis attend
to it or attenuate to it yeah of that
move movement to form the initially
false belief that I am
moving okay
okay yeah that's like just I don't know
something about that's very interesting
it's
a
so and then afterwards are you we are
trying to make up for that
by uh trying
to predict those uh or get those
appropriate like inferences right um
it's not you're not you're you're going
to start attending to it again or is it
you just decrease how much attention
you're putting on to those States at
first and then after
you're yeah
make certainly it's an oscillatory
pattern so one one analogy or another
case or sensory tenation is really
important is the ey Cade so when the
Eyes Are Fixed the Precision is is high
so that changes in the visual scene when
the Eyes Are Fixed can be informative

and then during head movement Andor eye movement you kind of have like a a hyper prior that pixel changes or or um are uninformative because they're they're not going to be very useful there's You're Expecting High volatility on when there's movement so during that there like a sensory attenuation during the seccade and then as soon as the Cade ends it's just like re-engages attention okay so it's kind of like when when the um when the signal to contract the bicep occurs there's also a descending relaxation of the tricep in in the kind of healthy setting so it's not just like pull on one side and pull on the other side it's like a joint contraction and relaxation signal that that's happening with two opponent muscles whereas the icade attenuation example is more like opponent processing with a motor and an attention variable but they they alternate okay so we have the isoc case with visual perception getting out of a chair well and there probably are other differences here like you could argue whether the iade cons it's kind of like it's considering the counterfactual um just like the getting out of the chair is I am not in a chair is the counterfactual the iade counterfactual is like I am looking over there but you're not

and so the the um distri the belief
distribution um ultimately you want to
shift your location in that
space But if the um current Precision
around the current state estimate
doesn't have significant Mass covering
that
counterfactual then it it can't be
switched over to so sensory attenuation
relaxes the
attention expands the variance reduces
the Precision makes it so it's like
instead of like 99% sure that it's this
way go well it's 80% sure that it's this
way 20% that it's that way that opens
the door to the counterfactual being
realized through action just to kind of
restated a bunch of ways because I
think if it is the case that that's how
um attention and attenuation work then
probably it's going to be prevalent in
like
every sensory motor
system because the alternative would be
constant
Precision yeah and then would so
constant Precision would make it would
actually make you less likely
to um be able to process movement
because there's a lot of
uh it's almost like you're be able like
less likely to get new information is
that kind of a way of thinking about it
is that why we're kind of relaxing the
attention so we could be more flexible
with sort of uh states that are changing
very
rapidly I think that's true so let's

think about what would happen if you had
constant Precision constant Precision to
your visual field in iade or constant
Precision to your proprioception for the
getting out of the chair case well when
your eyes were um
moving just where would you fix your
constant value if you fixed it at the
level of that that your surprise should
be at for being fixed then it'll be
shocking when there's movement if you
have it when it is if you um calibrate
to when the pixels are moving a lot then
you won't be able to glean as much
information when they're not moving
because you'll be dis attending or you
could pick a lukewarm value and kind of
blend those two be a little bit more
surprised than you should be when it's
surprising and then a little bit less
informative when your gaze is fixed so
obviously an approach to resolve this
tension is to have oscillation or some
kind of
conditional
attention so that you can tune it and do
the best you can do in each moment
instead of trying to like fix a
value when we already know that there's
a different
optimal Precision for
different um settings or like with
through time within the same
setting
okay
that makes sense it's kind of like
a uh trying to think of
a good sort of analogy but uh just like

you're actually doing worse by it's kind of like you know you're grasp in like have you ever heard of that analogy where you're grasping like a seed you can't hold too hard or it'll like slip out but if you hold it too loose it'll also slip out so you need to like adjust yeah that's that's good connection that's also related to kind of optimal grasp like Optimal grasp on a pen is not the maximum compress Force for for a variety of reason you know you'd be exhausted you couldn't have um dextrous movement or something like that so like then that's related to like not over under fitting and that's kind of the VF imperative which is you want to fit a distribution given the family of variational distributions that you're choosing from gaussian or whatever it is you want to parameterize it so that if you were to make it any tighter you'd be overfitting and then if you made it any looser you would be having left information that you could have used so it's on some trade-off Frontier um which we can be along with a KL Divergence so it's like a it's you know it's in a good spot and it's on a good optimizable manifold that if the variational family is at all appropriate you're going to get to the optimal grasp

now maybe it's like um trying to do optimal grasp on something that's too small or too large to be grasped that would be like if the distribution family was inappropriate then you'd You' come to the best grasp you could of you know the mountain but you there you it wouldn't be possible to actually do it so that kind of con actually like the the the model that doesn't under overfit still may have an adequacy of 1% given the data and all this other things MH and then and then one other connection there is like um the Colman filter or just generalized filtering so we're getting measurements from the thermometer Through Time one or more thermometers through time and then we're tracking the temperature distribution um or stock prices or something then you want to be tracking both the expected mean temperature and the variance like the volatility or just the uncertainty the the sort of fundamental uncertainty or volatility and then also the measurement Associated so the measurement Associated volatility is more like a matrix the actual Dynamics are more like B but and so but to whatever extent it comes the noise or the variability comes from either the sensor measurements or the underlying dynamics of the system you you want to be fitting this curve Through Time such that new noisy data

measurements are not like falling outside the curve because then that's more surprising than it should be then you should wide to the curve or um if all the data are falling like within the plus or minus one standard deviation on the curve then you could have done better by tightening the distribution son's work on the generalized filtering is really nice on this on the the generalized filtering kind of connecting into the the

Baye

um basing optimal filtering in perception and then bringing in action this also kind of reminds me of uh exploration exploitation tradeoff and reinforcement learning and uh soft value functions where they actually instead of just trying to optimize a reward they try to also increase the entropy like an entropy value um so which seems kind of counterintuitive but it's actually performs really well it makes it very flexible and able to deal with protuberances specifically so like disturbances like a if you have a robot that's trying to work in an environment where things can start disturbing it and things like that having sort of that like trying to optimize that entropy value as well as the reward value makes it so it can actually adjust and uh deal with disturbances a lot better which kind of reminds me of like movement also it's kind of like disturbances to sort of

your oh yeah yeah and uh there's two kinds of motor disturbances ones you have agency over and ones you don't have agency over which ones are more surprising the ones you don't have agency over like um you can um not a recommendation but you can slightly push the eye so that it's kind of like a non saccade induced visual shift of course that's surprising um or if uh you can't tickle yourself because the expected propriceptive consequences of tickling yourself are not attended to because they're expected so they're confirmed um or yeah just how we expect H how far do we expect the stair to be if it's a little bit more or less then it's surprising so then just the expected consequences of action are normalized too I mean Sensations are perceived calibrated to what their expected states are this is kind of the whole predictive processing everything is is is like sensory attenuation is kind of a

Cory

of um a first principles corollary of this kind of action and perception as inference and then it just like um uh predictive coding in the visual system this is empirically found in different systems and then and

then it'd be interesting like what other phenomena are like where do we see allostasis or attenuation in um in other systems like if you have um the bacteria and the gradient of sugar or like antibiotics or something it recalibrates to the value with like a kind of log sensitivity so that it can always have optimal grasp on making a decision about the gradient if the gradient were a thousand times longer than the bacteria it couldn't detect it if the gradient was a millionth of the wavelength of a bacteria it couldn't detect it but then there's some range that it can actually do the calibration that gives the most statistical resolution just like the aperture of a camera or the noise level of a sensor and so then when you're calibrating the microphone or the the whatever it is the two kind of two minimal things or the one minimal thing is just the mean that or that's like the Delta the dero Delta function is if you can only estimate one parameter about to calibrate the microphone in the room you would want to know the potentially the average or the mean of the median volume and then the second thing would be a variance parameter um those might be jointly

estimated for certain one parameter families of distributions like Poisson distribution where you only use one parameter to describe the family and then that is like where like the mean and the variance are equal is a one parameter estimation but in other case the kind of two that are helpful are the central tendency and then the variance it doesn't have to be gaussian but you kind of need those two because if you don't have a measure of this of a central tendency and then how wide a field of the central tendency then it's like you're kind of saying that it's a diverging situation so then it's not really identifiable good good stats pieces all of regular these are common topics and in other statistics settings and it's kind of cool because like they're often discussed in the signal processing inbound passive inference side common filter all the variational auto encoder and then also in in a kind of slightly siloed way in the control theory optimal control K optimal control way so then those two statistical problems can be understood as kind of like two

complimentary special cases there's
optimal signal processing when you have
no agency or no action control and then
there's optimal control in a completely
known

environment where there's no epistemic
value which is like the figure from
earlier with showing expected free
energy and then when certain things are
included or not what special cases
results so sensory

attenuation it's a
useful tractable

Motif that does

that that uh addresses the
the the kind of realistic statistical
problems by modulating and

[Music]

learning

attention another thing I was uh curious
about was the generalized synchrony um
where they're talking about the birds um
doing the bird song and then the the
other one would try to finish the other
segments of the song it just kind of
started going into uh something I'm like
uh been interested in which is like
communication and uh even like

Linguistics and

just like how that seems similar to
communicating language and like when you
talk to like a language model you're
like people a lot of people say it's
just predicting sort of the the best
response or what's the most likely next
response um and it kind of just remind
me reminded me of like this generalized
synchrony where the birds are singing

the song and then the next segment of
the song would be sort of it's kind of
like a response almost to the other bird
if there's like two birds singing
um yeah I guess I was just like Curious
in like how

that and communication works like is
this like the very most basic form of
communicating in active inference like
between two sorts of Agents
uh

like yeah yeah good
questions

Andrew oh yeah I just wanted
to I uh previously skimmed a paper by
friston and Par and they actually
reference it here at the bottom of
165 um it they built another model that
basically has two agents you know
communicating with one another and with
synchrony I I'd rather go with the bird
song example I guess since it's more top
of mine but um I it's just it's rather
fascinating there's this um described
like the birds are basically learning
one

another's model parameters right such
that they start making virtually the
same inferences you know one is with the
other um I think this is a wonderful
kind of example too because I can
imagine two musicians you know wanting
to play the same song staying in time
with one another kind of following the
same script or sheet music um but yeah
I'll I'll try to find that that paper
just because I think it's if you're if
you find interest in it it's you know

it's more based on potentially what humans would be doing um it includes the entire model of like how do you you know how do you construct um in communication like what is a question um what are the different states uh involved with questions asking one another questions answering questions and and like so I'd love to check that out yeah yeah that's cool yeah you even said like scripts and uh Alber Ro n all with the strong and weak scripts so you have two musicians it could be they could literally have the same sheet music and that could be truly isomorphic sheet music like they're both playing the first trumpet part on this one song the same time it could also be a general synchrony which is to say not necessarily a lock step maybe it's a call and response so there's still a perfect Mutual information between the two sheet music readings that are ongoing even though they might be playing different instruments or they might be playing different parts so that's like what in that bird song example and subsequently they've referred to as like the singing from the same hymn sheet and then kind of stepping out into the Improv is another step that I know some music and other researchers are are interested in because like with questions or other um turn-taking mechanisms because obviously it's different when there's epistemic transfer in communication of of whatever kind versus

just the

co- performance of something that's
fixed ahead of time so that's that's one
key piece and then um the uh the point
about the large language models is a is
a really good one so the the likeliest
continuation

of this trained model it's almost like
the current token is y That's the
observation and then we have like Q big
foundation model Q now it's not
necessarily a variational basian model
but then there's going to be some
variational free energy landscape and
then when the temperature is low it
picks simply the next
likeliest outcome

and then when the as the temperature
increases you flatten that landscape so
it starts sampling like more
broadly to to the point of breaking out
of kind of semantic ruts but decreasing
potentially like the fluency or the
coherence so that that's like we have
the path of least action and then we
have that temperature or shaky hand
parameter because in language selection
of topic and sentence and word and phon
name and all of that is like this nested
policy selection

task and then so we have both the um
path of least action that's kind of the
ridge on the bottom of the valley and
then we have this like sampling
technique where you can sample along the
strictest um likeliest path of
continuation or you can
sample more broadly like from that

Valley so it's an example of how like statistics or otive inference approaches could be taken even if GPT 4 whatever doesn't need to be applying active inference from the inside but we could still talk about its expectations or preferences and how that conditions its actions and all these different aspects but it's map not territory so we'll be doing like cognitive mapping of gp4 not not reverse engineering or engineering from the inside so the bird song example is a singing from the same hymn Sheet example where there's an expected and so it's an interesting and a good good uh continuation from the sensory attenuation they both have this an expectation in their gener each bird in in its generative model of how the song ised to go and then the question is are they going to be singing that song or not and so in this example it's like well like someone should be singing it this way and then it finds itself singing and then as it continues to sing takes its turn its um Precision drops we we could talk about why Precision might drop through time and then eventually that leads to a uh consideration hey maybe maybe I'm not the one continuing this song and then that's like the kind of turn-taking mechanic but it's not the

only turn-taking mechanic that you could design you could use like a stop you know word like over or something like that or start words or all you know you can you can make any mechanism and then I think they show in that first paper with frisen frii or in some other papers like maybe what one have you mentioned about like the learning of the model and in that case there was like a younger bird like a more plastic bird with low precision and then over multiple rounds it leared how the song was supposed to be and then the first bird was taken away and then the new one was added in and they showed that basically you could do like this successive passaging of entrainments in the generalized synchrony settle so they mentioned like this leads to the Birds trying to like match their internal models um by trying to match how they continue singing um and I was I mean I definitely want to check out like how it was like all of this was like how fren and all of them connected this to communication um because like I guess I'm like wondering like how you communicate bits of information uh directly because like it makes sense you match the internal models and then you uh I guess like you take turns and you communicate the next section and then I guess what it informs the other

bird just like from like a sort of based
like information Theory type
uh perspective it's like now the other
bird knows that they're singing the same
song um um I guess that is like some
information that's being communicated
and
then um yeah it's like matching the same
song and then I
guess uh I guess that is kind of like an
analog for language if you consider
every idea or concept a song and you're
trying to match with the same concept
cept or like some communication that
you're trying to
communicate I think there's some some I
was just gonna say I think there's some
some Theory somewhere I don't want to
misquote someone I think it was from a
neuroscientist
uh um but uh the idea that whenever two
people are communicating it seems to be
the case that including using like
neurophysiological measures like EEG or
otherwise that there tends to be an
increasing kind of synchrony between
them um that includes in the in the kind
of spectral Dimension like frequency um
like if you did like a spectral
decomposition of the EEG and look at the
frequencies going on like their their
frequencies tend to slowly begin to
match one another um and furthermore if
you kind of analyze the contents of the
conversation which I'm not quite sure
how you do this with high Precision
mathematically but um the idea is like
two people communicating with each other

are slowly reaching some kind of convergence or consensus in what the conversation is about um about agreements upon the topic even if you disagree at some point you do converge in the idea of well now I know your stance and I also know my stance so at least that kind of Truth has been converged upon so I guess to Summit up like communication is some kind of General uh process of of convergence or syy of of information of brain frequencies um and I guess I guess here with um chapter eight uh synchron synchronization of chaos because I guess we haven't really gotten into that but like using the Lauren system as one potential like continuous model as well sharing all that that reminds me of like how communic a in uh like Wolf's ruad is sort of like between observers is kind of explained it was like uh between The Observers they have like shared like segments of the ruad um that they could both like kind of agree upon um which kind of reminds me of sort of like trying to match your internal model or you know like what you were saying like at least you converge that you both or or some understanding even if it's on an agreement it's like you're converging on how your stance is related to the other person's stance where like that relationship itself is kind of like a concept on its own um which is very interesting it's like we each we we're putting both of our

cards on the table now we can see each other's cards so to speak like that information has not been shared um right even if the information that you brought to the table uh was different uh right now you've kind of come together here you have shared information and furthermore you are in the an instance of a particular conversation right so it's yeah

yeah and like in information Theory like communication channels um it's kind of like um you're trying to it I mean a lot of this chapter reminded me a lot of communication channels from information Theory just like trying to know what the original input was that goes across a channel and then you get an output but the output might not be the same as the input and you have to see if it matches and if it's if there's noise then there might be some variation with the output so it's just kind of like trying to decrease that noise and then that's why you calculate the capacity of what's sort of the max rate of bits you could communicate across this channel without sort of um where you know for a fact that um you'll be able to map it directly to sort of the original input or like the

cause so yeah a ton of great connections I think it's gonna be really exciting to see the quantum

FP Chris Fields semantic information flow meet up with wolf from ruad I think that should be impending and but um yeah and then there's kind of like the the

agent environment communication that's
Alice and
Bob and then sometimes we can kind of
take the um Alice as agent environment
as agent Bob as agent and then just sort
of like summarize it with just a
holographic screen with the classical
information which is like saying that
there's an
inert uh intermediating entity so
Quantum on either side of the boundary
and classical or n dimensional on the
boundary and then n plus dimensions on
the in the
quantum

[Music]

um and then there's uh simpler examples
too like having a bunch of metronomes on
a table so if the table is
fixed and then there can't be like force
transduction amongst the
metronomes they will continue unabated
with whatever population distribution is
initiated and then if there is coupling
nonzero coupling then like that's the
kind of famous videos where they enter
into lockstep they're simple or
conservative active
things so the general synchrony that
they come to is like
either all aligned or a two-phase but
then I've also seen it where it's
two-phase and then it kind of Falls a
little bit off and then it actually
slides all the way to back to lock step
because that's a slightly more
favorable synchrony but they're all
exact versus one where say it's like

5050 or 991 with one going in the counter phase so that's kind of like the simplest generalized synchrony and then it's a huge open topic with semantics and Linguistics seems like there's a lot of information in the sequences of tokens how many bits how many knots relative to what but also it's not just the sequence of tokens we know that in our generation of language and our seemingly our perception that of course the timing tone context all these other features come into play so then what is their Mutual information and there's settings where maybe just the tokens are necessary and sufficient there's another situation where the Tok there might not even be a token like a silent sign what does that say so then yeah these are great questions of communication I think we'll continue to see like in a lot of different fronts like ratcheting of the okay now sophisticated and the affective and and then they have these models of the nested look at this this is like figure 4.3 identical on top and then the bottom whereas in figure 4.3 in the textbook the bottom is the continuous time GM here we see the discrete time GM converted into the for Factor graph and that supports the message passing more approximately but but we're dealing with this exact same discrete

time model and then as we see with like the folk psychology models often the more cognitive or symbolic tasks are discrete and then the hybrid model is like where it hits the ground it's continuous like for example um the larynx um contraction or voice box or lip or tongue might be a continuous time continuous actuation model so you have some continuous um um voice apparatus with a propri reception and the actuation making phonemes but then once you're in the space of identifiable phonemes um or lexicon then you can be at this level where they're now making discrete time models at the symbolic so they've abstracted away from the phon they're not dealing with that here I'm saying that's that's kind of implicitly underneath it they're just saying that that they discretely produce and recognize these words but then that's what's so nice about the composability of ACM okay now here they're taking it deep we see that four3 but now there's each time step has three nested so that's discrete with Indiscrete nesting and then you could also um just say instead of the observation being a discrete token you could plug in a continuous model there and then you can evaluate you know empirically or what are the

computational requirements or where is
it where you getting more um value
at and I think well the chaos thing
that's explored a lot in the uh
stochastic chaos and marov blankets
paper I think 34 live stream
34 and then it's interesting how like
even though a lot and I think Thomas par
may have remarked on this in the book
stream 2.1
like the earliest ACM models were the
continuous
time and then to
accommodate classically symbolic
cognitive processes and also explicit
planning as inference
and also to to integrate with a
PDP well known in planning there was a
movement towards the
discret and then there's theost at all
2020 like a synthesis of active
inference in discreet State spaces so
not like it's like finished and people
continue to develop new motifs and
discrete compositions but kind of like
the fundamentals of the
discret were then focused on more and
then potentially now with the path-based
formulation of basian
mechanics G Theory everything that
Dalton and Maxwell at all are working
on there's kind of like a a a a
renewed emphasis and generalization on
continuous structures like paths sheath
bundles
topos things that the book book doesn't
necessarily go into but sets the stage
for and then figure 8.6 kind of like

um shows in the simplest possible example which is the same one that live stream 46 the ACM does not contradict folk psychology the simplest hybrid model where you have a discreet higher level symbolic um Dimension and then this kind of like sensory motor continuous time so this is like figure 4.3 get smooshed together and then yeah I mean just in terms of the location in the book second half is the more pragmatic uh model you know construction and implementation based Focus starting with chapter six with the recipe um and then just like the low and the high road were sort of this dialectic that that are unified or contained you know of or by active inference in seven and 8 we have that distinction and then again with with this figure 8.6 by the end of chapter eight we're already landed with the synthesis of seven and eight whereas two and three they really are like different branches and they meet in the middle in seven and eight it's more like I mean they're just tucked together at the end of eight and then chapter nine introduces okay well what if instead of specifying a GM hasht chapter 6 and then generating synthetic data what if we already had collected data of a certain structure we knew that we were going to collect data of a certain structure how

could we then prepare a GM that would
recognize that data allow us to do
statistics on it whether we do discret
continuous or Hybrid models chapter nine
and then the end of the
book so it's just like so funny how
second half starts fast
describes a few key here's the
vertebrates here's the invertebrates or
or whatever uh here's the kingdom of the
fungi here's the other
Kingdom then chapter nine gets more
empirically
oriented and chapter
10 simply reviews and contextualizes
active
inference so it's a quick
ride it is very fast
yeah let's just see what the other
questions were how is five figure 52
related to figure 84
52 hierarchical predictive processing
84 luren
synchronization we should we should
procedurally generate every single
possible question how is everyone every
every figure I
mean um I mean for a for a brief second
I thought maybe it would related it to 8
figure 8.1 which is a little more I
don't know um not quite message passing
but at least it's you know we're still
in the realm of looking at error
terms basically these these basic
equations with
Lauren's to go with Lauren's systems
though and relating that to message
passing um I'm not sure I'm not so sure

there's no clear there's no yeah no
where my brain quickly
goes well that's the
attenuated space where any account could
be possible or more accounts could be
possible
like again to this kind of question
prompt question answer or even hegelian
problem reaction solution structure it's
interesting like which questions just
have an
answer which ones are like the Grand
Canyon where the continuation there
should only be like one specific
continuation like what's the
relationship between policy and
affordance there may be a formal
answer not even to say that that would
be the only answer still
but there might be a Sumo
expressable necessary sufficient answer
which is like pretty good but then for
for for any question but especially some
of
them these questions are basically just
evocative which is totally fine I think
it's
fun we're only going to experience even
just with a 30 or 50 or how many figures
there are I mean that would be like
200 questions
MH so so just like then so that's the
epistemic foraging which question should
we
ask prompt
engineering conversation
Theory all these different things
like folding back into

our the way that we're learning
active like okay now that we've looked
at what are the causes and consequences
of sensory
attenuation now what are we going to pay
more or less attention to or now that
we've talked about communication along
these
lines then how do we do
what lot of fun options
there here's the improvisation
question yeah
I'll just move it to this
one
any any uh last
thoughts um yeah so in chapter nine is
it more about like using active
inference on sorts of like if you wanted
to use it on like a data set to detect
something or you know things like
that um yeah basically it points that
way it is not the guide book or the
Google collab notebook or whatever that
just oh yeah load in your CSV and then
this is just going to analyze the data
for you so it's not to that point but
what I think nine does that's important
first
9.1 is a schema of the behavioral
scientific setting where you have a
behavioral science laboratory making a
cognitive model of the rat or the ant or
whatever which might be a map back of
the environment but it frames it in a
way that we can handle with ACM and
then analogous to how chapter six gave a
recipe for making the GM figure
92 gives us the road map from going from

the data plus the
GM to basically the kind of figure that
would be in a published
paper like we proposed um a personality
variable or a psychometric variable of
this kind continuous or discrete or
whatever it
was we had the generative model we had
these data and likelihood function we
inverted the model in light of our prior
beliefs which might be uniform across
groups or how whatever whatever it was
in the setting and we think and this is
kind of like Ryan Smith's Precision
Psychiatry work you'll see a ton of
things like this it's like and people
who had this or this categorization or
this or that continuous
variable here was the correlation or the
regression between our inference about
this parameter and whether or not they
were diagnosed this way and here's the T
test and it um looks like something that
you'd find in a paper in other words
supporting an empirical claim being part
of a fabric of evidence about an
empirical
system rather than just having a
hypothetical
GM and then speculating like it could be
this way or it could be that
wayc I mean and until you parameterize
it and especially fit of data because it
usually
could so kind of like a six and little
bit of a Jimmy Hendrick
thing okay all
right thank you fellas see you

um next time see you next week bye bye

thank you byebye